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Letter of Recommendation of *Hyeonbeen* (*Edward*) *Lee*

Dear Admission Committee,

It is my great pleasure to write this letter in strong support of Hyeonbeen (Edward) Lee and his application to your doctoral program. I have worked closely with Hyeonbeen (Edward) Lee for several years as a research collaborator and co-author on two research papers[1,2], and on a funded research project with the Korea Atomic Energy Research Institute (KAERI) [3]. These collaborations have allowed me to evaluate his academic ability, research skills, and personal qualities in depth, and I can recommend him to you without reservation.

Our collaboration began when Hyeonbeen (Edward) Lee joined the KAERI project[3], whose objective was to develop and validate a reliable simulation model for a spent nuclear fuel transportation system. Within this project, he played a central role in the work that led to the research paper[1], taking responsibility for understanding the physical system, constructing and refining a multibody dynamics model, and tuning model parameters to match experimental test conditions. This work demonstrated his ability to connect engineering practice with rigorous analysis, manage a complex modeling workflow, and communicate effectively with researchers at a national laboratory.

Building on this background, Hyeonbeen (Edward) Lee led a line of research on physics-aware learning for impulsive dynamical systems, culminating in a first-author publication in the Journal of Computational Physics and an Excellent Paper Award from the Korean Society of Mechanical Engineers. Starting from the insight that standard learning models struggled with higher-order quantities such as contact accelerations, while smoother, lower-order dynamics (positions, velocities) were more amenable to learning, he formulated a principled strategy that organized the learning process around these stable quantities and validated it through focused experiments. This project illustrates his ability to move from physical intuition to a well-posed research question, an implementable method, and a thoroughly validated result—an essential quality for doctoral research.

Through our collaboration, I have found Hyeonbeen (Edward) Lee to possess a rare combination

[1] S. Han et al., Multi-body dynamics model for spent nuclear fuel transportation system under normal transport test conditions, *Nuclear Engineering and Technology*, 2023, 55(11), 4125-4133.

[2] H. Lee et al., cNN-DP: Composite neural network with differential propagation for impulsive nonlinear dynamics, *Journal of Computational Physics*, 2024, 112578.

[3] Development of a Simulation Analysis Model for Land and Sea Transport Tests Using Multi-Body Dynamics and a DNN-Based Meta-Model, *Korea Atomic Energy Research Institute* (*KAERI*), 2022.

of strong analytical skills, solid engineering judgment, and broad intellectual curiosity. He has a firm grasp of core concepts in dynamics and machine learning and is able to integrate them in a consistent and mature way. In project meetings he articulates assumptions clearly, questions modeling choices when appropriate, and is equally comfortable discussing physical reasoning and algorithmic considerations.

In terms of work habits, temperament, and interpersonal qualities, Hyeonbeen (Edward) Lee is very well suited to the demands of a rigorous PhD program. Our collaborative projects often involved tight deadlines, complex software infrastructures, and unexpected obstacles, which he consistently handled with calm persistence and a systematic mindset; when experiments or models did not perform as expected, he diagnosed the issues carefully, revised the setup where necessary, and improved the methodology rather than seeking shortcuts. On a personal level, he is mature, respectful, and highly collegial, working well with peers and junior students, accepting critical feedback constructively, and, over the course of our collaboration, showing clear growth in confidence and independence as he increasingly takes the initiative in proposing improvements and preparing research presentations.

In summary, I consider Hyeonbeen (Edward) Lee to be an outstanding candidate for doctoral study. Through our joint works, he has demonstrated strong analytical and programming skills, sound engineering judgment, and the ability to drive research projects from conception to validated outcomes. He has a clear and well-motivated research direction and the personal maturity required to thrive in a demanding international environment. I therefore recommend him to your PhD program with the highest level of confidence and without reservation, and I encourage you to consider him for any fellowships or financial support for which he may be eligible. Please feel free to contact me if additional information would be helpful.

Sincerely,

Seongji Han, Ph.D.,
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